PYTHON FROM FIRST PRINCIPLES

Beginner Examples Collection

1. VARIABLES & DATA TYPES

```
name = 'Subham'
age = 27
height = 5.9
is_student = True

print('=== VARIABLES & DATA TYPES ===')
print(name)
print(age)
print(height)
print(is_student)

print(type(name))
print(type(age))
print(type(height))
print(type(is_student))
```

2. OPERATORS

```
a = 10
b = 5

print('Addition:', a + b)
print('Subtraction:', a - b)
print('Multiplication:', a * b)
print('Division:', a / b)

print('Greater Than:', a > b)
print('Equal:', a == b)

print(True and False)
print(True or False)
```

3. CONDITIONS

```
temperature = 35

if temperature > 30:
print('Hot Weather')
else:
```

```
print('Cool Weather')
```

4. LOOPS

```
# FOR LOOP
for i in range(5):
print(i)

# WHILE LOOP
x = 0
while x < 5:
print(x)
x += 1
```

5. FUNCTIONS

```
def add(a, b):
return a + b

result = add(5, 3)
print('Result:', result)
```

6. DATA STRUCTURES

```
# LIST
numbers = [1, 2, 3, 4]
print(numbers)
print(numbers[0])

# DICTIONARY
student = {
'name': 'Subham',
'age': 27
}
print(student)
print(student['name'])
```

7. INPUT / OUTPUT

```
# Uncomment to test user input
# user_name = input("Enter your name: ")
# print("Hello", user_name)

print('=== INPUT / OUTPUT ===')
```

8. MEMORY REFERENCES

```
a = [1, 2, 3]
b = a

print(a)
print(b)

b.append(4)

print('After Modification')
print(a)
print(b)
```

9. OBJECT ORIENTED PROGRAMMING (OOP)

```
class Car:

def __init__(self, brand):
self.brand = brand

def start(self):
print(self.brand, 'Engine Started')

mycar = Car('Toyota')
mycar.start()
```

10. SIMPLE CALCULATOR PROJECT

```
def calculator():

x = 10
y = 5

print('Addition:', x + y)
print('Subtraction:', x - y)
print('Multiplication:', x * y)
print('Division:', x / y)

calculator()
```

=== PROGRAM FINISHED ===